

STRETCH Lab Project Report

Waviness Quantification of Collagen Fibers from SHG Images

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Timeline: Jan-May, 2022

Background

Ligaments are connective tissues that support internal organs and hold bones together while collagen fibers are their building blocks having great tensile strength. In this project we specifically study the uterosacral ligaments (USL) that are important supporting structure to support the uterus and apical vagina within the pelvis. Our focus is on the waviness of the collagen fibers in USL, as it determines the maximum strain that the tissue can sustain as evident by the toe region of the non-linear stress-strain curve of such fibers (Figure 1). We use the straightness and orientation of fibers to characterize their waviness tracing the SHG images of Gilt tissue (Figure 2). We also study the limitations of the software (CT-FIRE) that was used for the tracing to identify possible scope of improvement in such quantification.

Methods

Analysis Tool Used: We use CT-FIRE, a MATLAB supported software that provides tools to automatically extract individual collagen fibers from images for quantitative assessment. The algorithm of CT-FIRE includes a de-noising step (i.e. curvelet filter) applied on the original FIRE algorithm developed by Stein et al. (2008) for extracting 3D biopolymer network architecture.

Imaging Technique: Second Harmonic Generation (SHG), a second-order non-linear optical process was used for the imaging of the collagen fibers. In this process, two photons interact with the material and are reconverted to form a new emitted photon with halved wavelength and doubled energy (Figure 3).

Results

Waviness Quantification: 20 samples of SHG images of collagen fibers taken at different depths of a segment of Gilt tissue are analyzed. We report the results in terms of the straightness and orientation of the extracted fibers. We present the statistics tracing an

individual image (Figure 4, 5) and also the mean statistics averaged over all of the 20 images (Figure 6).

CT-FIRE Limitations: We scrutinize the effectiveness of CT-FIRE using a set of drawn curves with varying width, waviness, and opacity (Figure 7). It is clearly evident that the failure rate of the software increases with increasing waviness, non-uniform width, non-uniform opacity. Failure is decided when a long single curve is traced as a set of broken short segments marked by different colors.

Conclusion

- Mean straightness of the fibers is 91% with a standard deviation of 8%. Orientation of the fibers is skewed toward 0° and 180° , meaning that they are mostly aligned along in-vivo loading direction.
- Long wavy fibers with varying thickness and brightness in the images are often wrongly captured as multiple smaller segments by the software we used, which can be improved further.

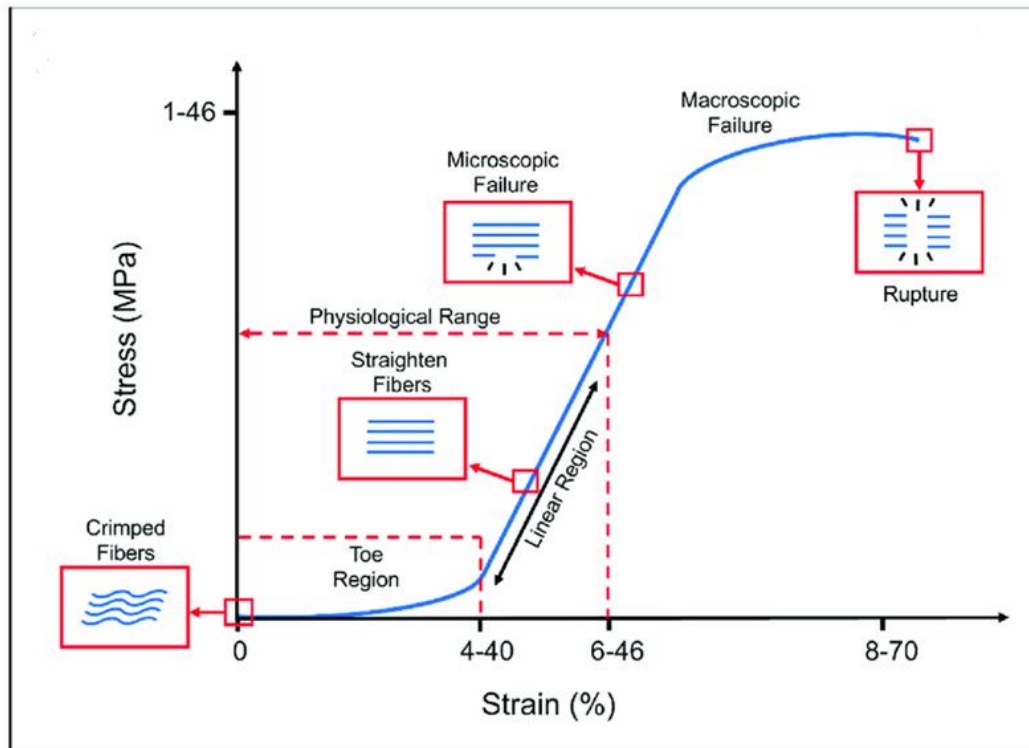


Figure 1: Typical stress-strain curve and the behavior of the collagen fibers for ligaments

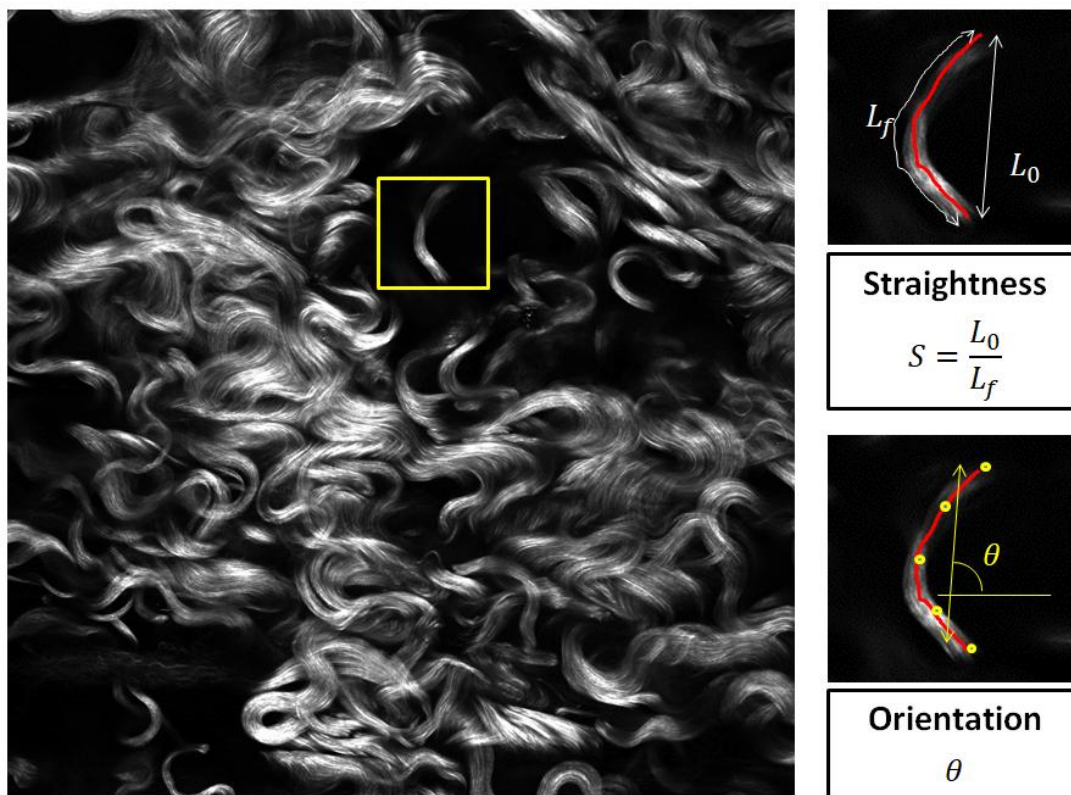


Figure 2: Sample SHG image and explanation of the parameters involved in the quantification

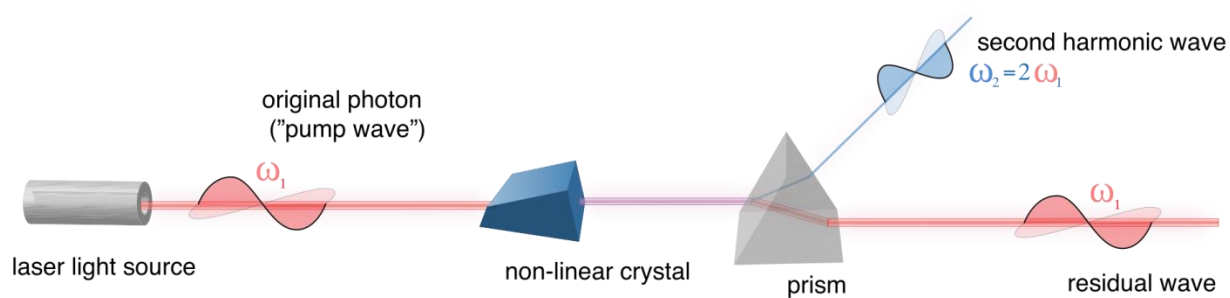


Figure 3: Schematic diagram of the second-harmonic generation process

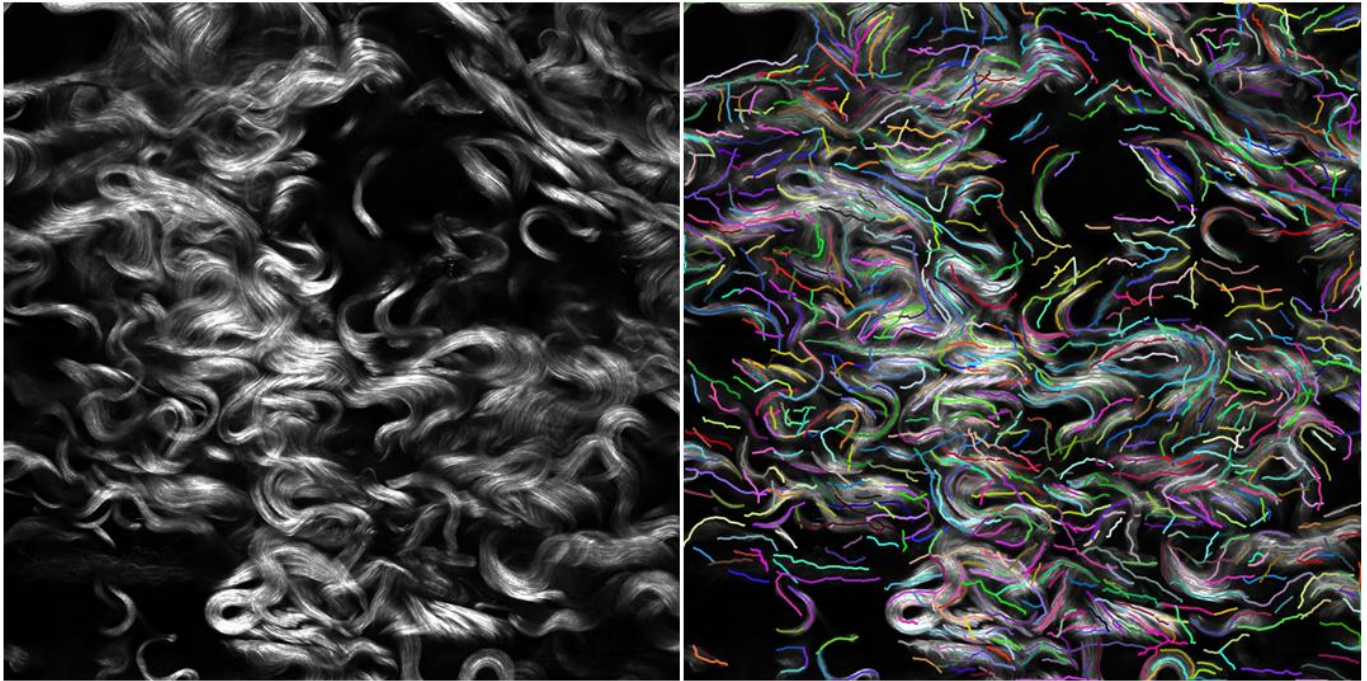


Figure 4: Traced fibers overlaid on sample image no. 15

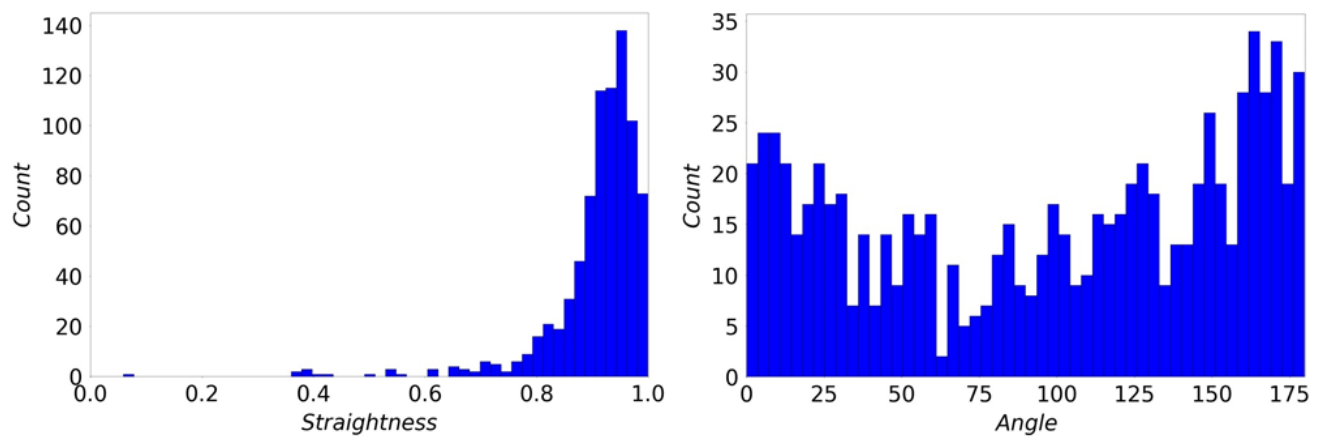


Figure 5: Straightness and orientation statistics for sample image 15

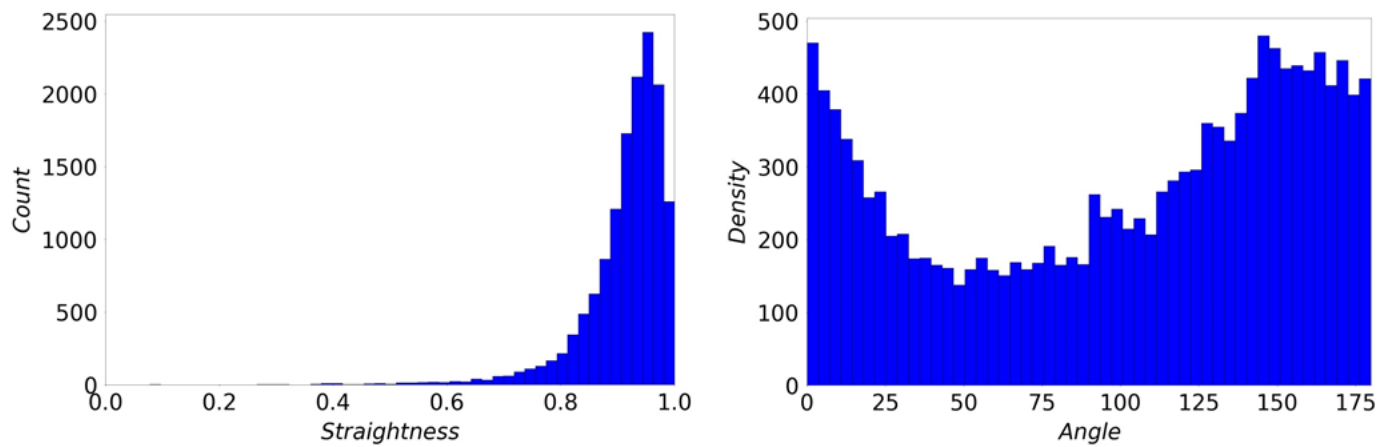


Figure 6: Straightness and orientation statistics over 20 sample images

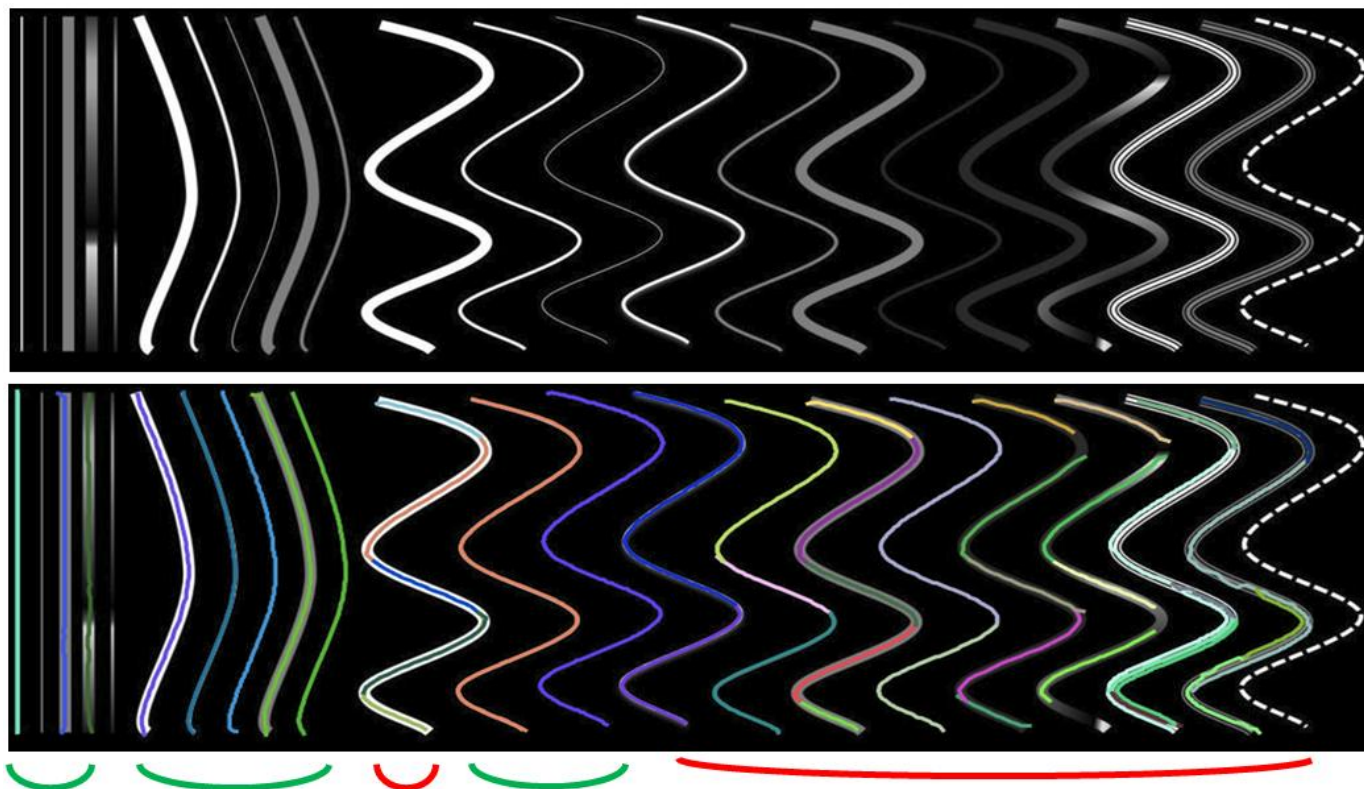


Figure 7: Effectiveness check for CT-FIRE using curves with varying width, waviness, and opacity